

Digital Twins for Verification and Testing of On-Orbit Robot Services

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Service Robotics, but in the Low Earth Orbit (LEO)

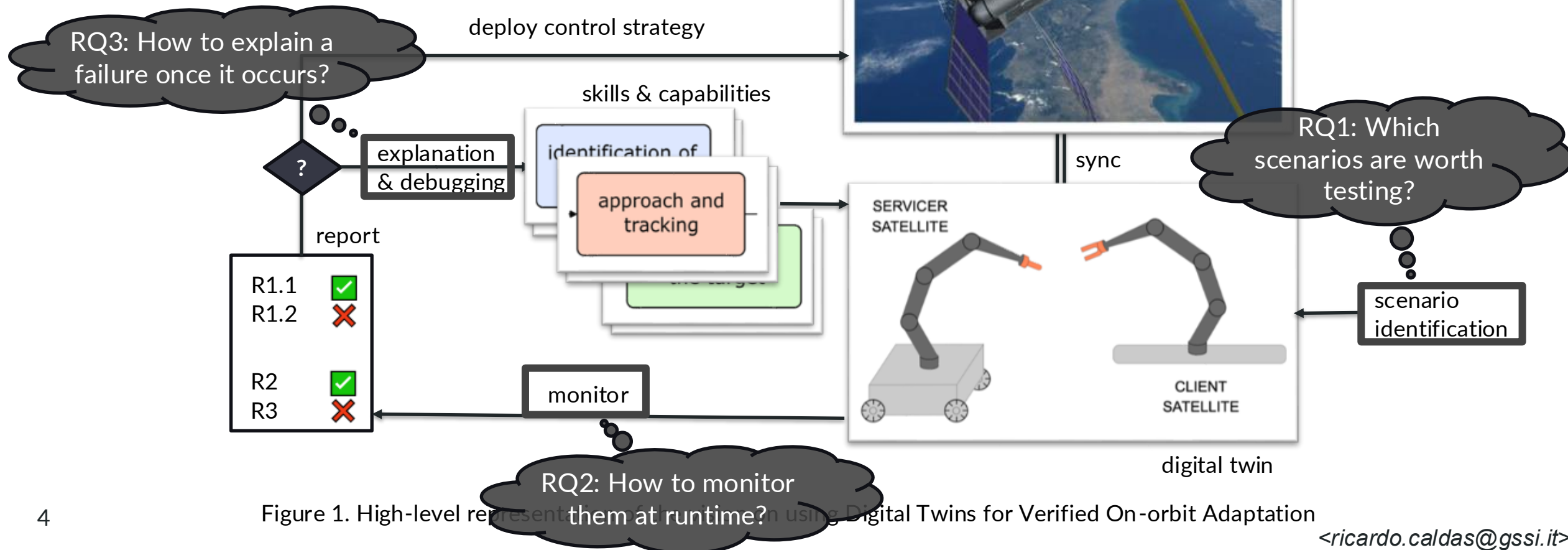
- LEO is congested, and disposal is now a regulatory obligation.
- Human-in-the-loop is unfeasible due to maneuvering dynamics.
- Satellites or Debris may be uncooperative.



Service Robotics, but in the Low Earth Orbit (LEO)

Capturing an **uncooperative**, possibly **tumbling** body with unknown inertia properties **requires a manipulator** with force control and **real-time autonomous maneuvering**

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Architecture & Requirements

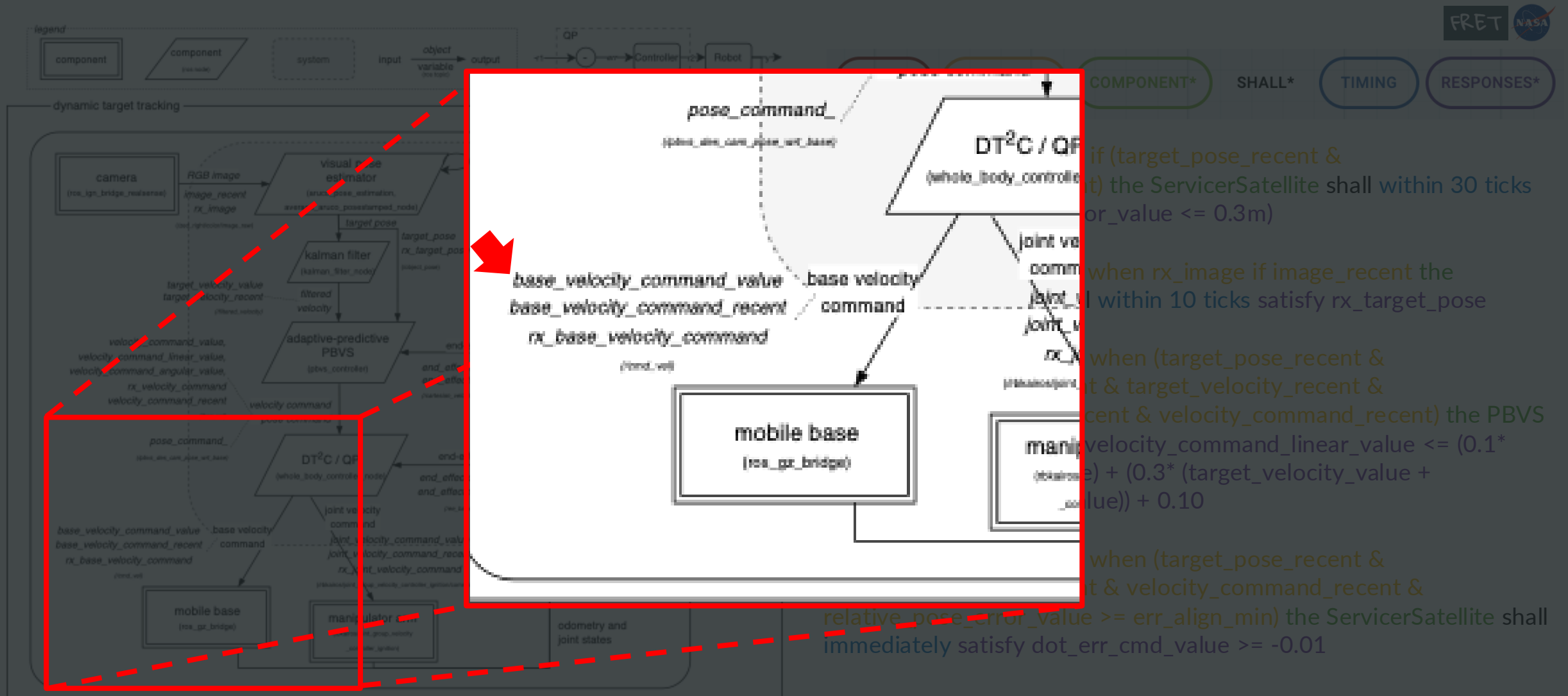


[fig. architecture]

[req. fretish]

Figure 2. ADDL architectural representation of the dynamic tracking controllers

Architecture & Requirements



Architecture & Requirements

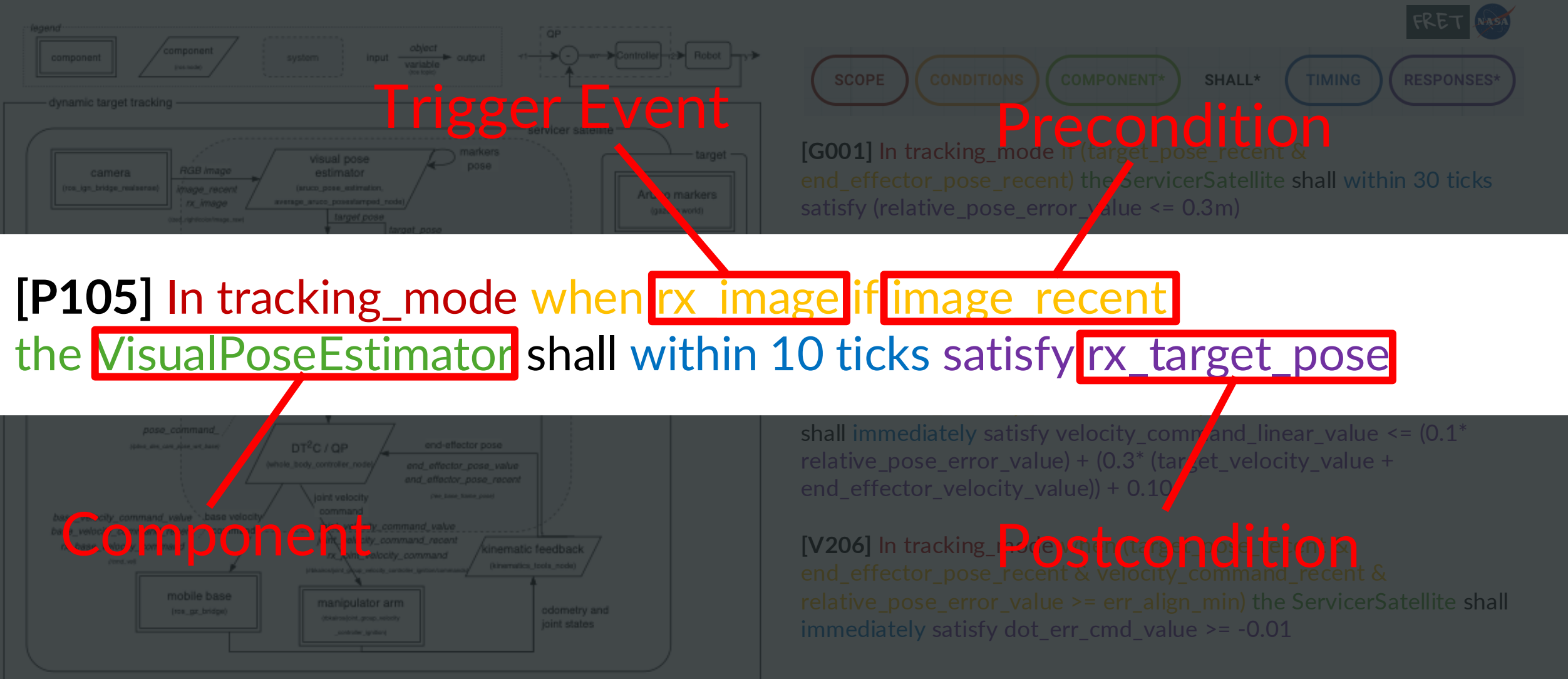
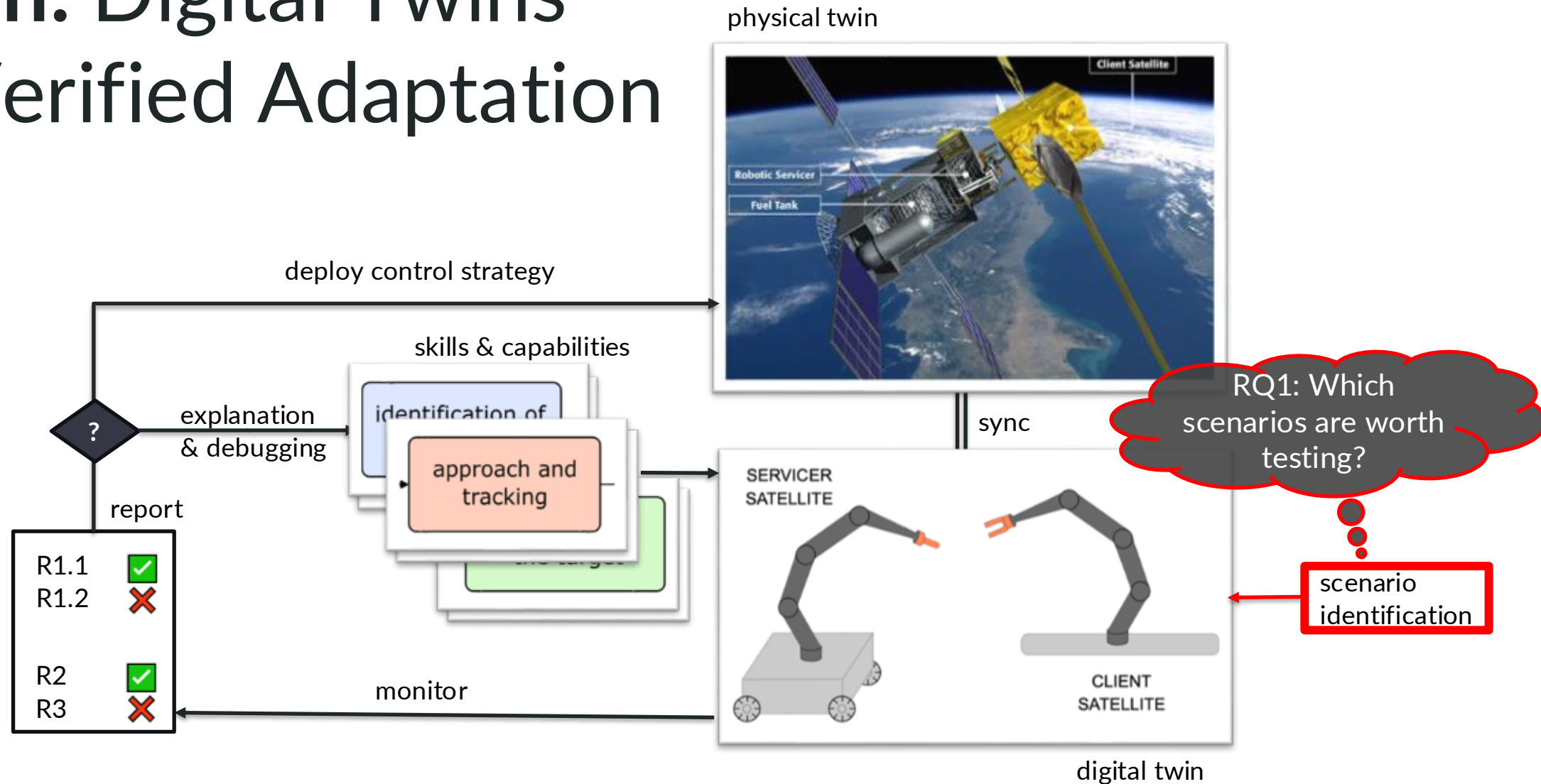


Figure 2. ADDL architectural representation of the dynamic tracking controllers

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Finding Scenarios via Obstacle Identification

- FRETish to Goal Models (flat->hierarchy)
- Creativity of LLMs + Rigorousness of Solvers (z3)
- Loop (DT & OS) until no new obstructions

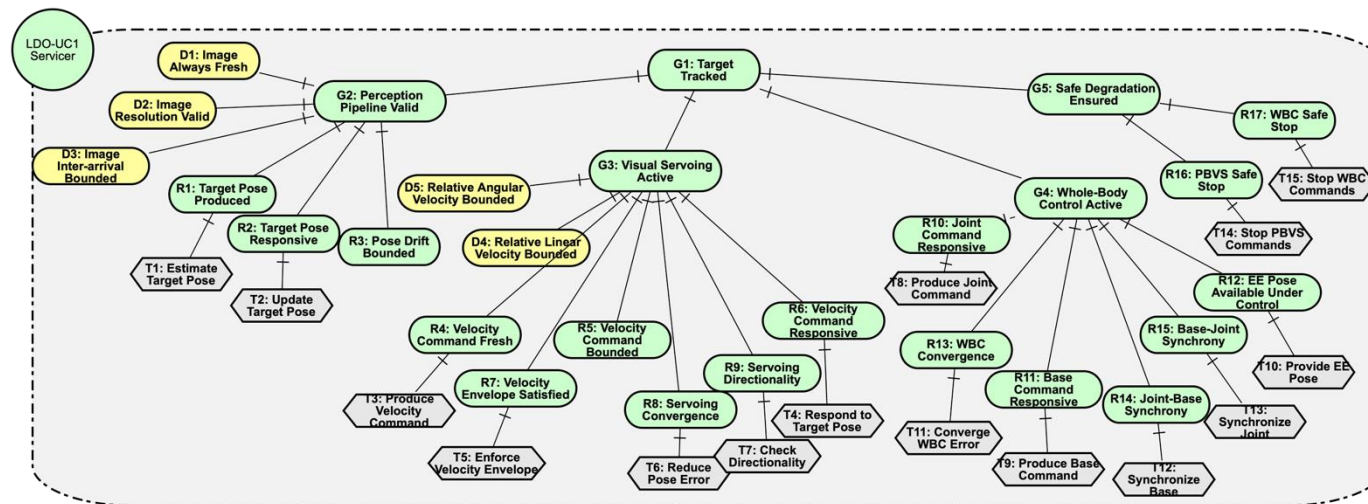


Figure 3. Goal model representing the Low-orbit objectives and operationalizations

O is an Obstacle.

O obstructs G iff ::

$$Dom \wedge O \models \neg G$$

O is feasible iff ::

$$Dom \wedge O \neq false$$

O is not trivial iff ::

$$Dom \wedge \neg G \wedge O \neq false$$

O is complete iff ::

$$(\neg O_1 \wedge \dots \wedge \neg O_N) \wedge Dom \models G$$

Van Lamsweerde, Axel, and Emmanuel Letier. "Handling obstacles in goal-oriented requirements engineering." *IEEE Transactions on software engineering* 26.10 (2000): 978-1005.

9 Letier, Emmanuel, and Axel Van Lamsweerde. "Obstacle analysis in requirements engineering: Retrospective and emerging challenges." *IEEE Transactions on Software Engineering* 51.3 (2025): 795-801.

Alrajeh, Dalal, et al. "Generating obstacle conditions for requirements completeness." *2012 34th International Conference on Software Engineering (ICSE)*. IEEE, 2012.

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Scenario Identification with OiK

OiK: Goal-oriented Obstacle Identification Toolkit

Human

- Domain Theory Elicitation
- Domain Theory Review

OiK method

- Domain Theory Derive (z3)
- Domain Theory Check (z3)
- Write Prompt (z3)
- Inject & Check Obstacles (z3)

Language model

- Obstacle Synthesis

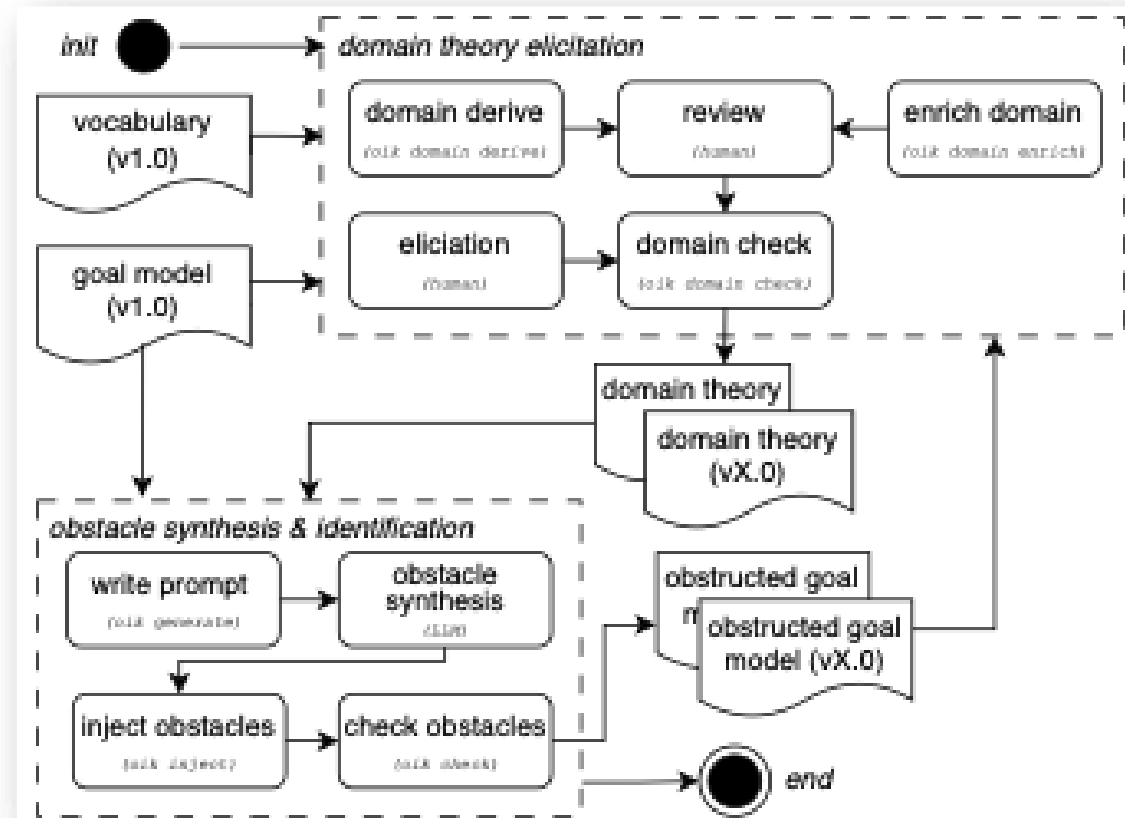


Figure 4. The OiK method

Produced Obstacles and Scenario Candidates

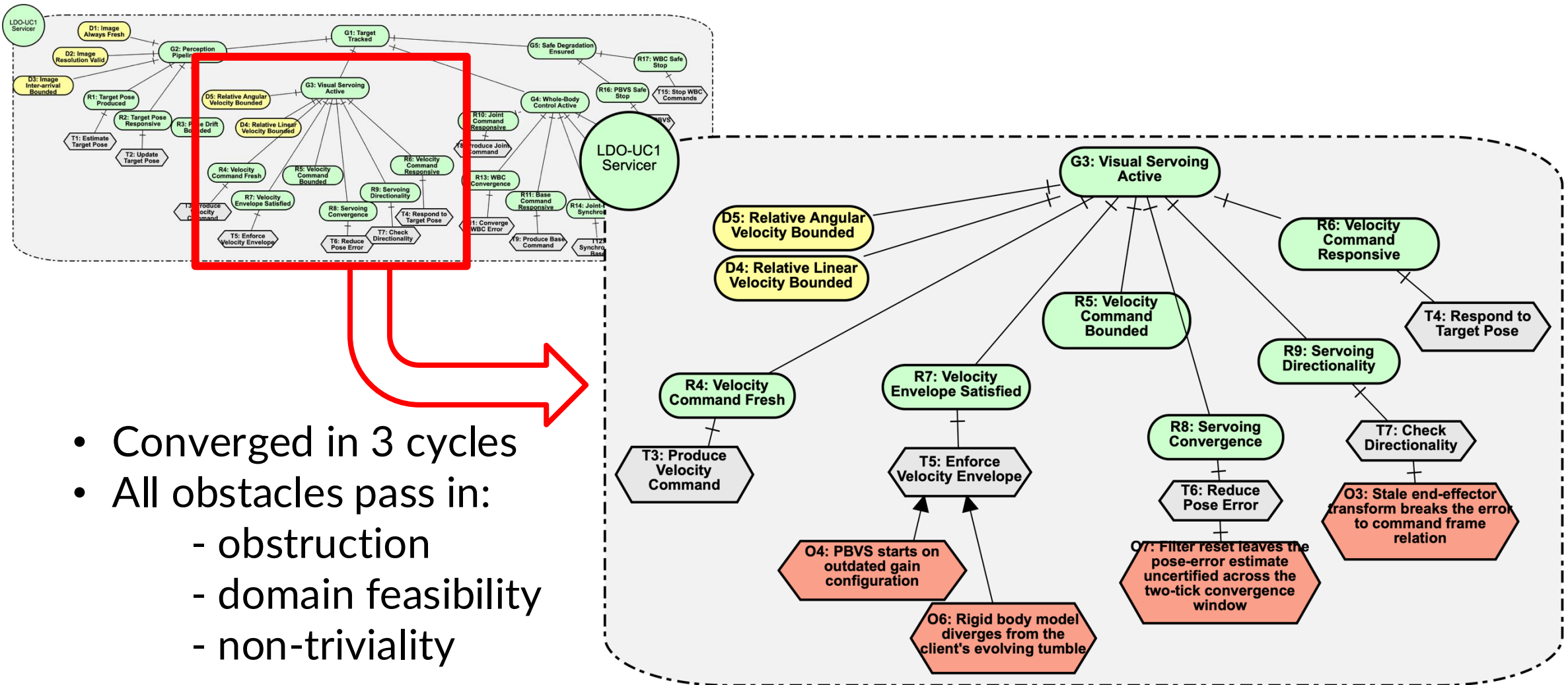


Figure 5. Obstructed visual servoing goal

Obstacle O4: PBVS starts on outdated gain config.

O4: (`compute_velocity_command` and $\{target_pose_recent\}$ and $\{end_effector_pose_recent\}$ and $\{target_velocity_recent\}$ and $\{end_effector_velocity_recent\}$ and $\{velocity_command_recent\}$ and (`not` $\{pbvs_gains_configured\}$))

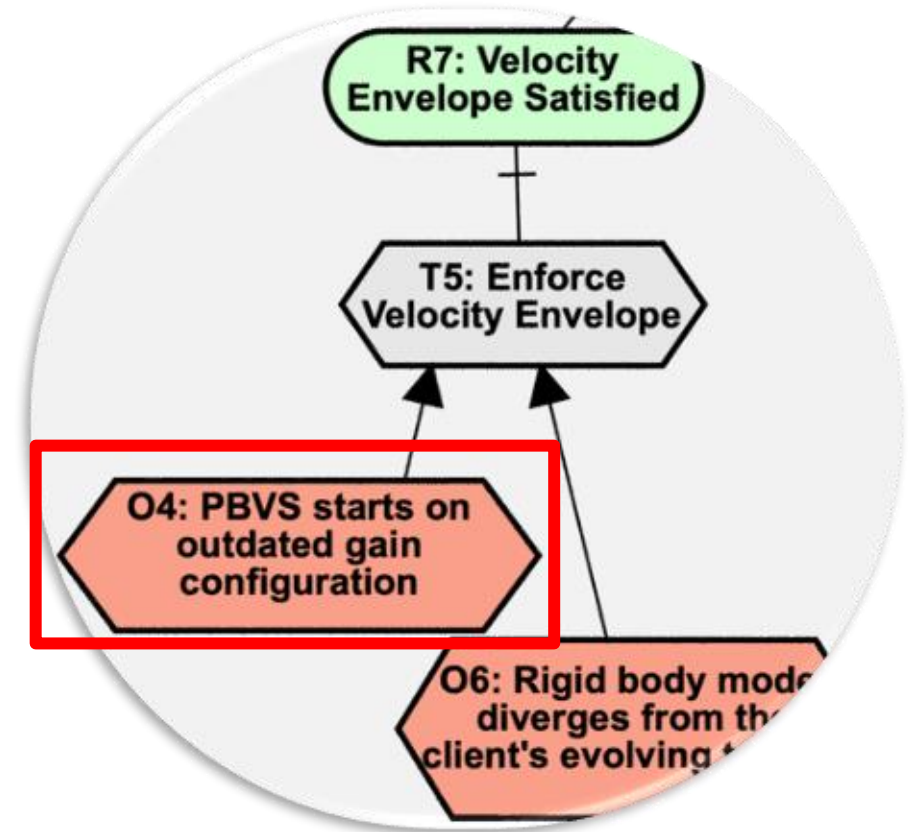
Proved from A7: ($\{velocity_command_linear_value\} \leq \{pbvs_lin_envelope_bound\}$) $\rightarrow \{pbvs_gains_configured\}$

[G3] In tracking_mode the PBVS shall always satisfy `velocity_command_recent`

Obstacle Class: Wrong Resource

Uncertainty Source: Execution context (Environment)

Severity: Defeats G3, G1 (Critical)



O4 Scenario: Outdated High Feed-forward Gain

[VIDEO]

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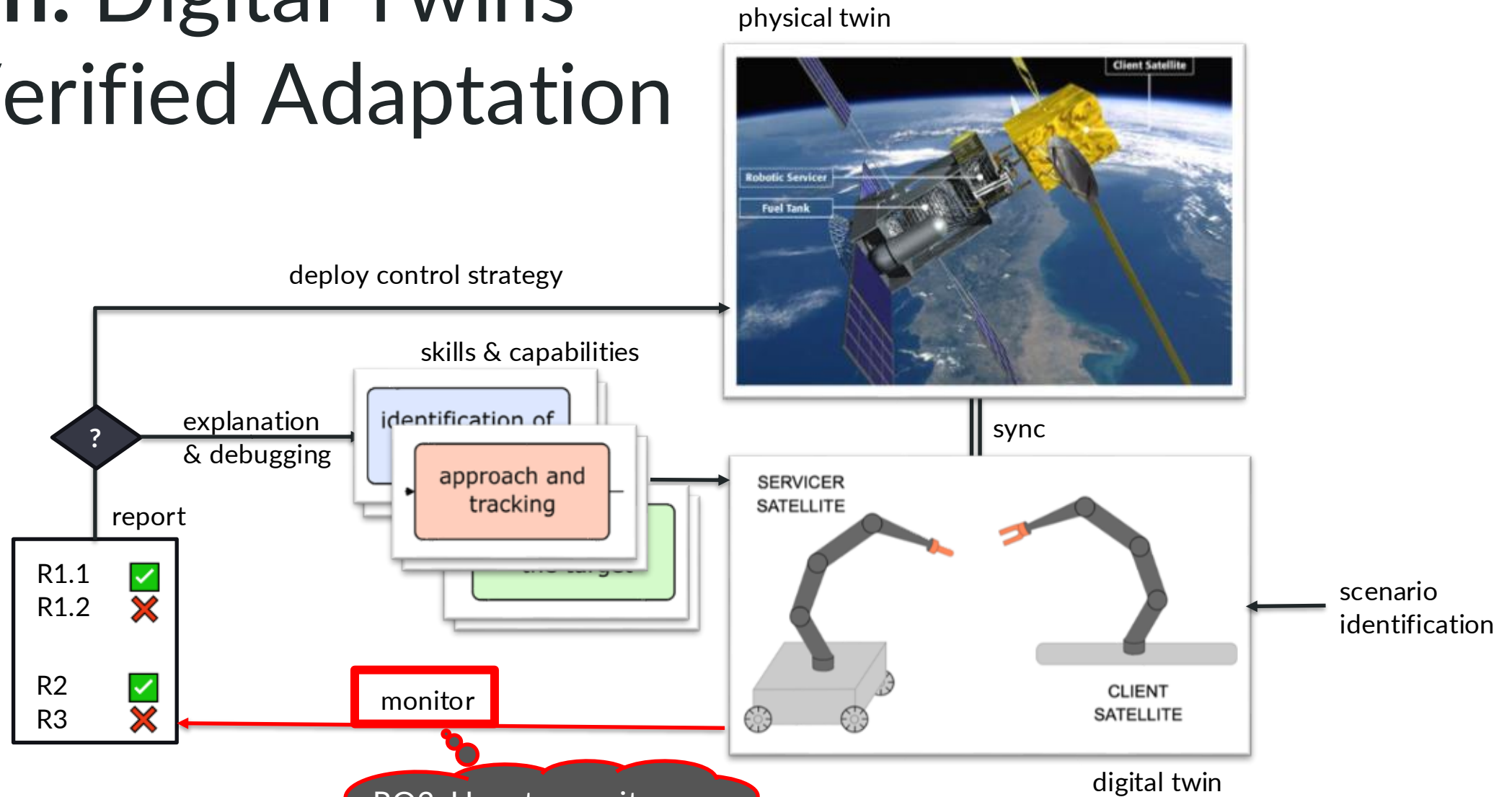
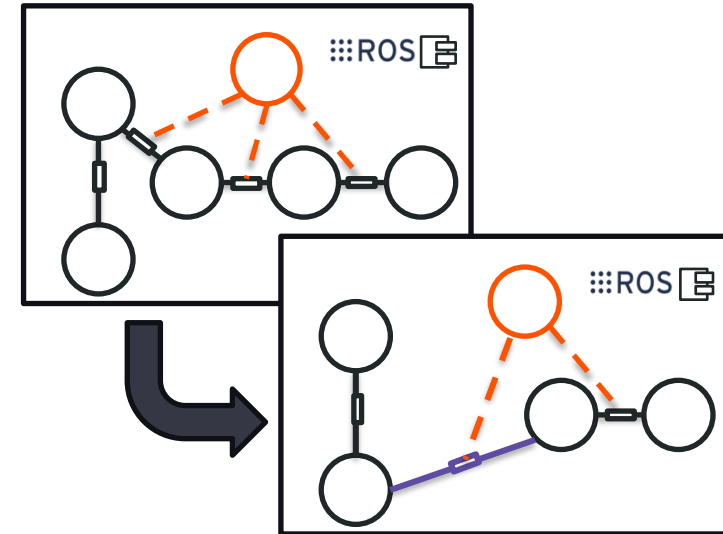


Figure 1. High-level representation of a physical twin and its digital twin for verified adaptation

Runtime Verification with RiTMOS

RiTMOS: Runtime Verification with Model Adaptation

- ROS topic semantics
- Past-time Metric Temporal Logic
- *Adaptable:*
 - *node/topic re-configuration*
 - *requirement change*



When target_pose_recent the PBVS shall
within 5 ticks satisfy `vel_command > 10`



When target_pose_recent the PBVS shall
within 5 ticks satisfy `vel_command > 10`
&& `ang_vel_command < 1.5`

Stream-based Runtime Verification

Monitors synthesis

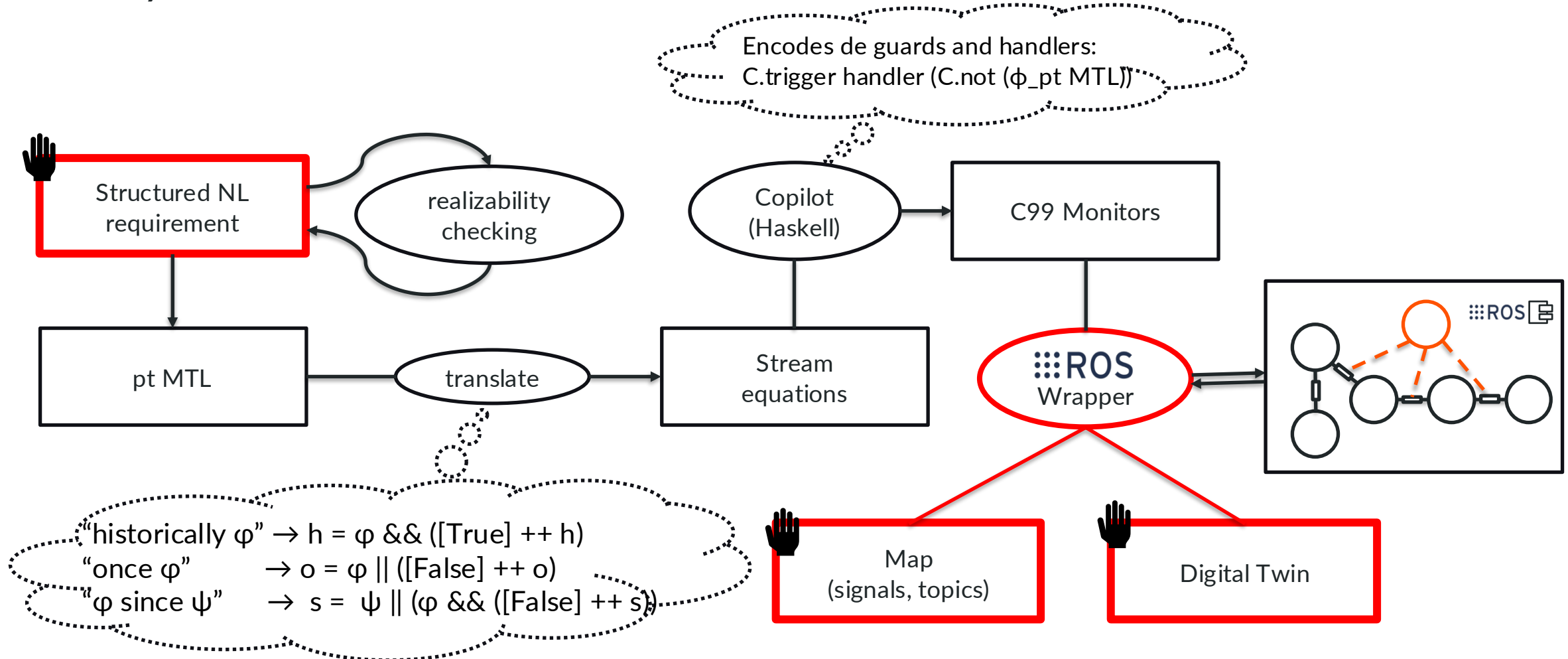


Figure 6. ROS monitors synthesis process following the stream-based RV approach

Preliminary Results: effectiveness and efficiency

- Preliminary experiments:
 - High fast-forward gain scenario
 - We deployed 5 monitors (V206, V205, V204, V202, P103)
 - Observed larger lag on V206 (~50ms)

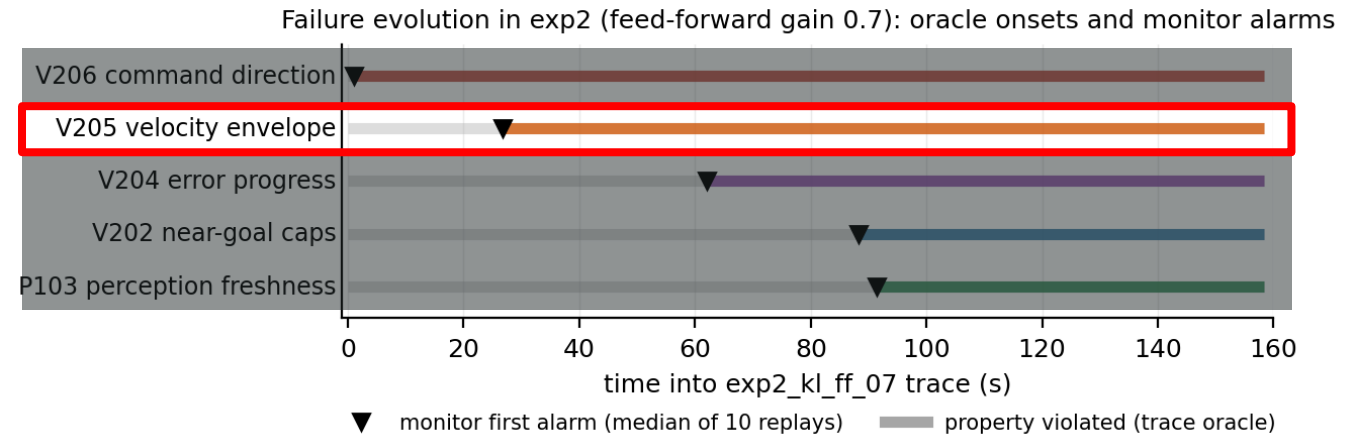


Figure 7. Effectiveness evaluation of deployed monitors

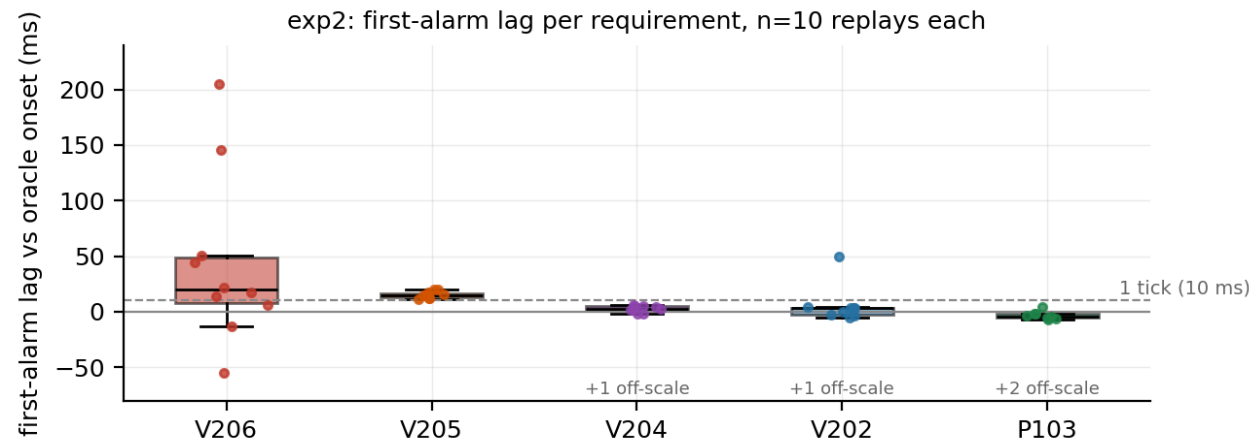
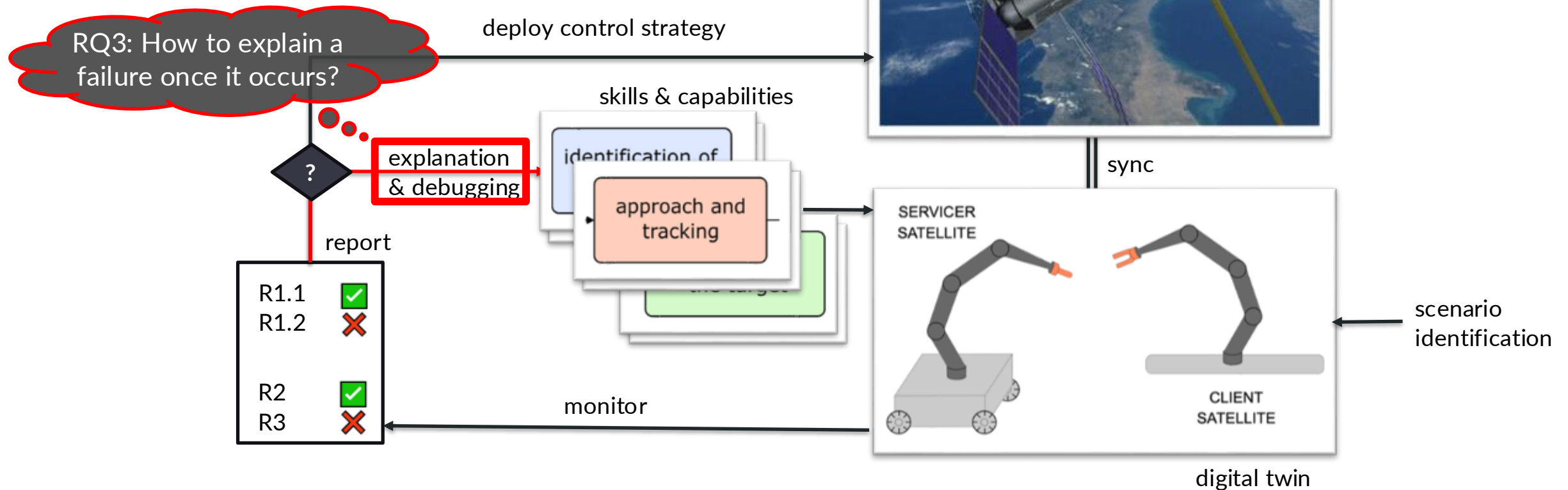


Figure 8. Efficiency evaluation of the deployed monitors

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Explanation and Debugging

Why did my system fail?!

[V205] In tracking_mode

when (target_pose_recent & end_effector_pose_recent &
target_velocity_recent & end_effector_velocity_recent &
velocity_command_recent)

the PBVS shall immediately

satisfy

$$\begin{aligned} \text{velocity_command_linear_value} \leq & \\ & (0.3 * \text{relative_pose_error_value}) + \\ & (0.5 * (\text{target_velocity_value})) + 0.10 \end{aligned}$$

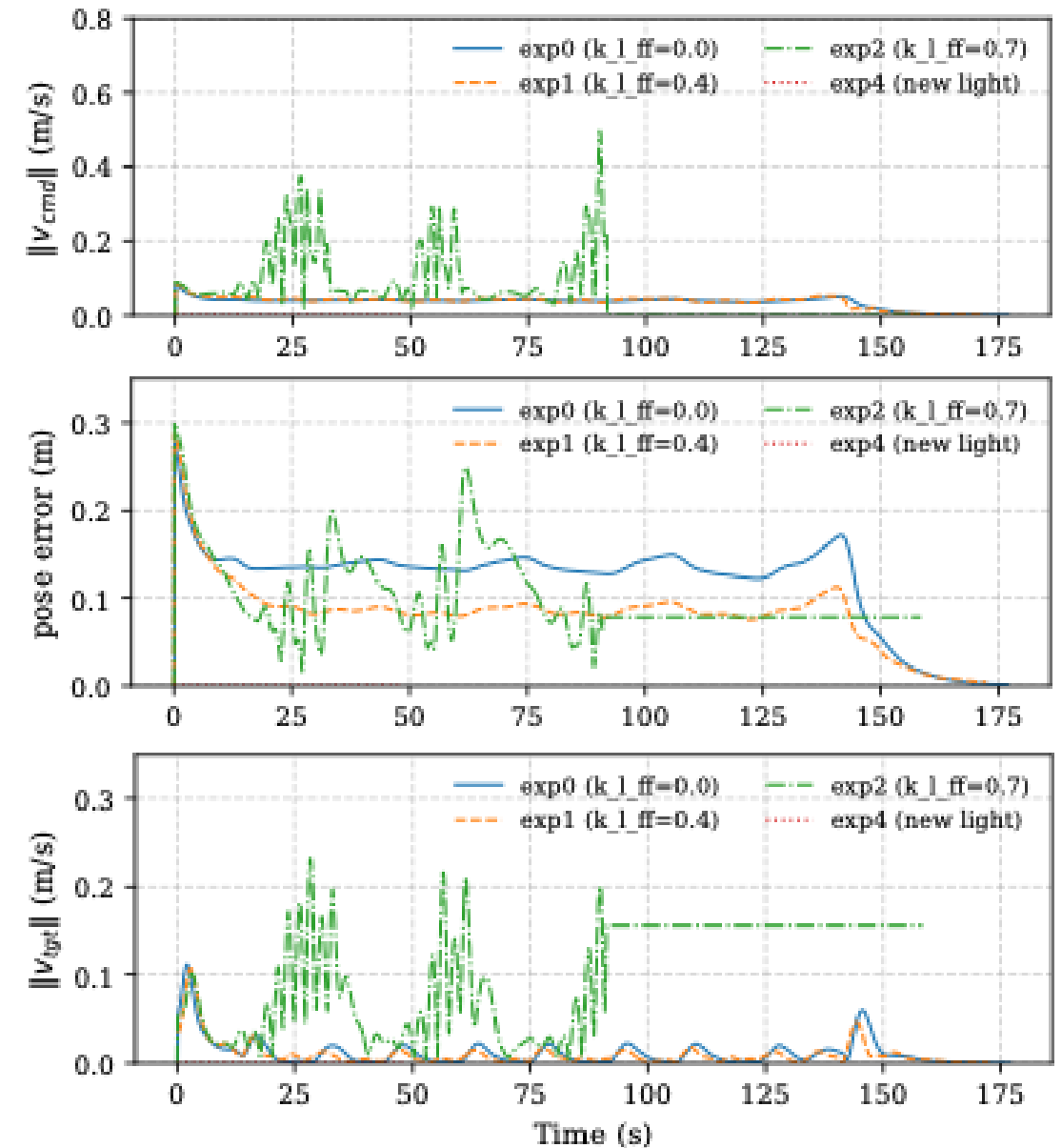


Figure 9. Execution traces for the scenario exp2

Search-based Trace Diagnostic

- Change-driven Diagnosis
- Genetic Algorithm
- Trace-Checking

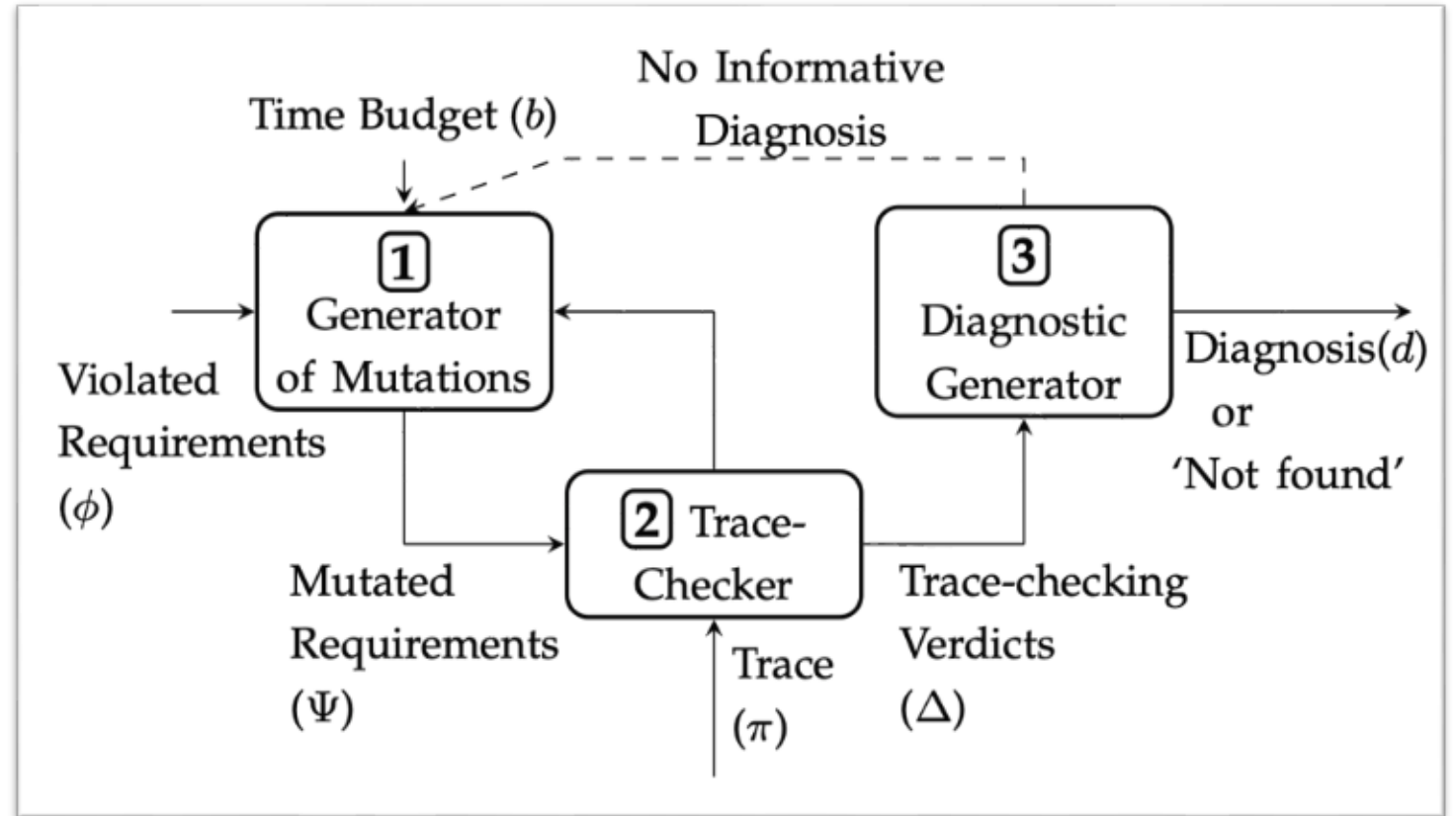
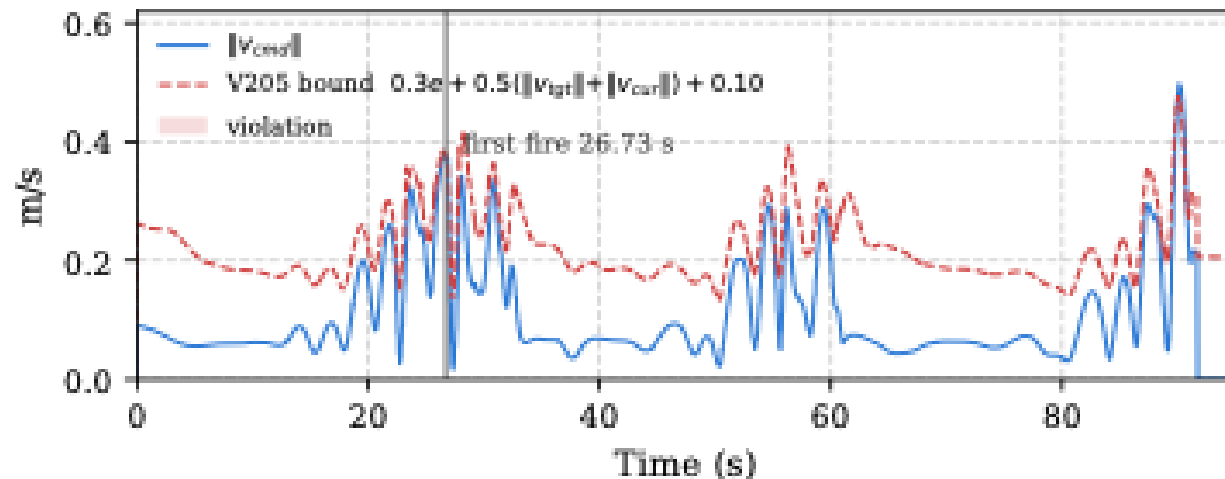


Figure 10. The Search-based Trace Diagnostic (SBTD) method

Diagnosis and the Diagnostic

- The high FF gain (0.503) yields the highest information gain, but it is **not the only configuration that leads to failure!**
- This provides valuable information on how control engineers can maintain high gain and adjust the other coefficients as needed.



$$\text{satisfy } v_cmd \leq (0.3 * \text{pose_err}) + (0.5 * (v_tgt + v_curr)) + 0.10$$

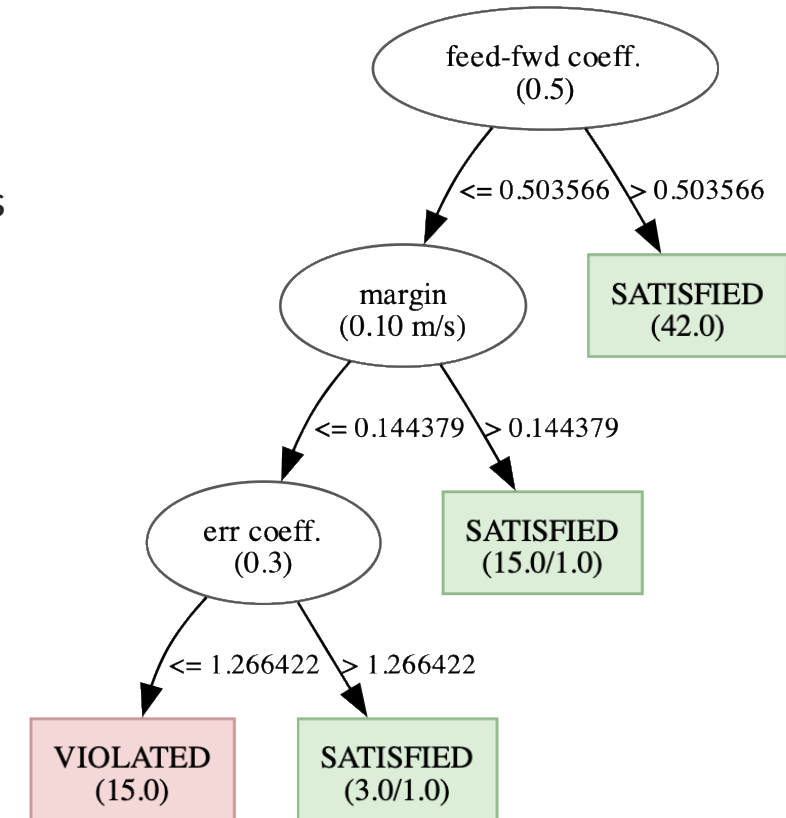


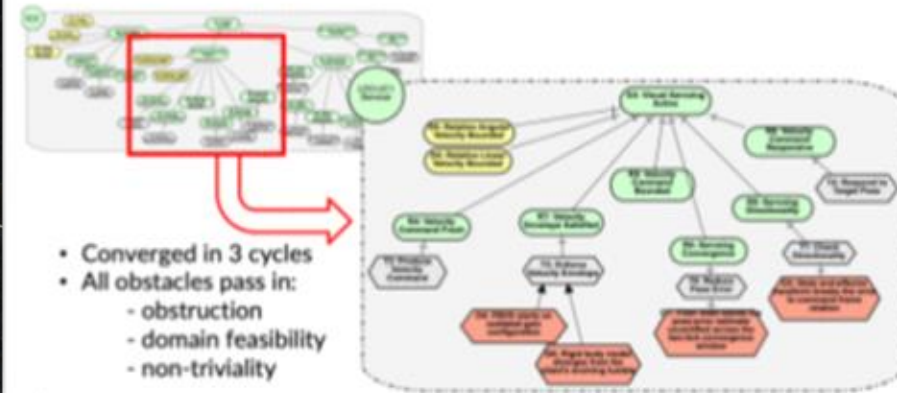
Figure 11. Decision tree encoding the diagnostic for exp2.

Wrap-up

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Produced Obstacles and Scenario Candidates



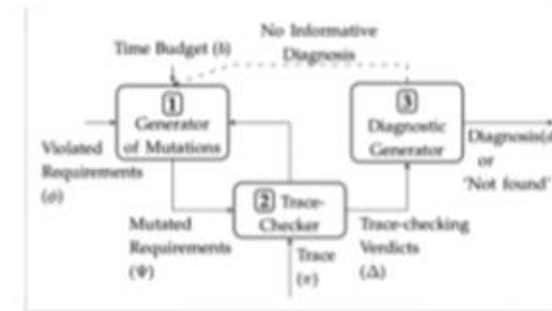
Stream-based Runtime Verification

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Search-based Trace Diagnostic

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Thanks for your attention!



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